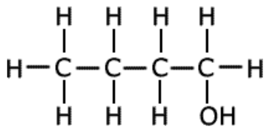
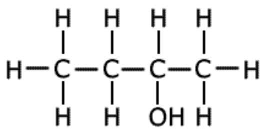
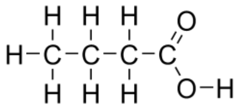
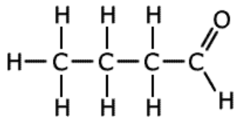
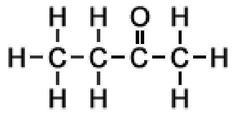
Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks available					
11.	*(a)*	(i)		Labels on diagram to show vertical condenser (unsealed at top) (1) Water in and out of condenser in correct direction (1) Heat source below flask with reagents (1)	3			3		3
		(ii)		Liquid evaporates, vapour goes into condenser, cools and goes back to liquid / condenses	1			1		1
		(iii)		Any of following for (1) • The reaction is slow • Allows time for equilibrium to be established • Stops reactants / products boiling away			1	1		1
		(iv)		Catalyst/ dehydrating agent	1			1		
	(b)	(i)		Fractional distillation (1)	1			1		1
		(ii)		Moles of ethanoic acid = 0.05 and moles of methanol = 0.04 (1) Theoretical yield of methyl ethanoate = $0.04 \times 74 = 2.96$ g (1) % of theoretical yield = $\frac{1.18}{2.96} \times 100 = 40$ % (1) ecf possible		3		3	3	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		Reflux for longer (1) Reaction is slow/ needs time to establish equilibrium (1) or Add extra methanol / ethanoic acid (1) To allow more of the acid to react/ push the equilibrium to RHS (1)			2	2		2
	(c)	(i)		Dehydration/ elimination	1			1		
		(ii)		Displayed formulae of butan-1-ol and butan-2-ol for (1) each  butan-1-ol  butan-2-ol		2		2		
		(iii)		Orange to green	1			1		
		(iv)		Oxidation of alcohol / redox	1			1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(v)		Displayed formula of butanoic acid / butanal / butanone   		1		1		
				Question 11 total	**9**	**6**	**3**	**18**	**3**	**8**